

Session : Evening Time : 2:00 pm to 5:00 pm

Total Marks: 70

[10]

- ④

(P.T.O.)

- Q2. Answer the following short questions (Attempt any TEN) [20]**
1. List different types of binding. Also give the sequence in which different binding are performed.
 2. What is TLB? State its purpose.
 3. Draw figure representing dynamic relocation with the help of a relocation register which uses MMU.
 4. What is demand paging?
 5. What is Belady's anomaly?
 6. List implementation methods for LRU page replacement algorithm. Explain any one in brief.
 7. What is a boot control block and volume control block.
 8. Draw diagram of typical FCB.
 9. List different memory allocation methods.
 10. List various I/O port registers.
 11. State the conflicting trends of I/O device technology.
 12. What is double buffering.
- Q3.a. Explain paging model in detail. [06]**
- Q3.b. Explain in detail swapping process in brief. [04]**
- OR**
- Q3.a. What is fragmentation? Explain external fragmentation in detail. [06]**
- Q3.b. Explain concept of segmentation in detail. [04]**
- Q4. Explain FIFO page replacement algorithm in brief with its disadvantage. [10]**
Calculate the total number of page faults for given reference string using FIFO algorithm (No. of frames = 3) :
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0
- OR**
- Q4. List different page replacement algorithms. Also Explain steps involved in a page fault with diagram. [10]**
- Q5.a. Explain layered structure of file system with figure. [05]**
- Q5.b. Explain linked allocation method in detail. [05]**
- OR**
- Q5.a. Write short note on Directory implementation. [05]**
- Q5.b. Explain indexed allocation method in detail. [05]**
- Q6.a. Write a detailed note on interrupt driven I/O Cycle. [06]**
- b. Write short note on Backup and recovery. [04]**
- OR**
- Q6.a. Explain in detail direct memory access with steps in DMA transfer? [06]**
- b. Explain in brief the basic functions of hardware clock, timers and cache memory. [04]**